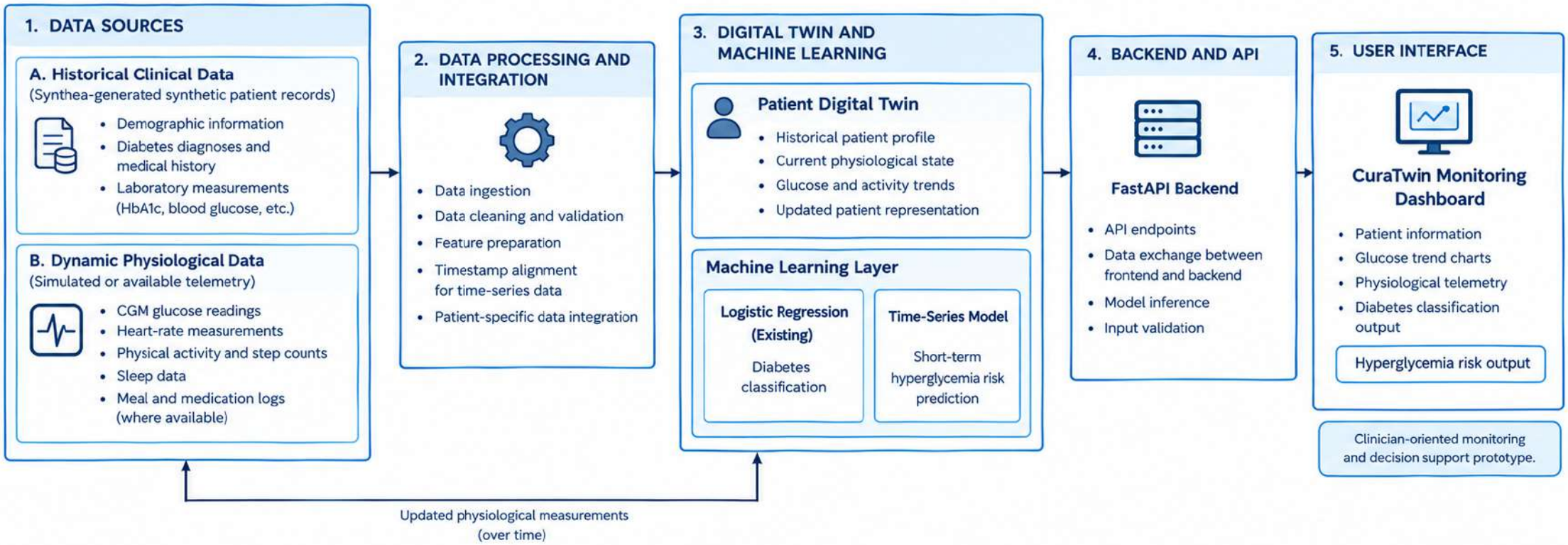

1. DATA SOURCES
A. Historical Clinical Data
(Synthea-generated synthetic patient records)
Demographic information
Diabetes diagnoses and medical history
Laboratory measurements (HbA1c, blood glucose, etc.)
B. Dynamic Physiological Data
(Simulated or available telemetry)
CGM glucose readings
Heart-rate measurements
Physical activity and step counts
Sleep data
Meal and medication logs (where available)
2. DATA PROCESSING AND INTEGRATION
Data ingestion
Data cleaning and validation
Feature preparation
Timestamp alignment for time-series data
Patient-specific data integration
3. DIGITAL TWIN AND MACHINE LEARNING
Patient Digital Twin
Historical patient profile
Current physiological state
Glucose and activity trends
Updated patient representation
Machine Learning Layer
Logistic Regression (Existing)
Diabetes classification
Time-Series Model
Short-term hyperglycemia risk prediction
4. BACKEND AND API
FastAPI Backend
API endpoints
Data exchange between frontend and backend
Model inference
Input validation
5. USER INTERFACE
CuraTwin Monitoring Dashboard
Patient information
Glucose trend charts
Physiological telemetry
Diabetes classification output
Hyperglycemia risk output
Clinician-oriented monitoring and decision support prototype.
Updated physiological measurements (over time)